EXPERIMENT 2: OSCILLOSCOPE AND FUNCTION GENERATOR

INTRODUCTION

Digital oscilloscope and function generator

A digital oscilloscope is a complex electronic device composed of various software and electronic hardware modules that work together to capture, process, display and store data that represents the signals of interest. An oscilloscope typically displays a given variable as a function of another variable. The most typical use case is the display of voltage versus time, where nearly all phenomena can be converted to electrical signals by using an appropriate converter.

Figure 1 below illustrates the generic block diagram of a digital oscilloscope. The front end's job is to scale and filter the analog signal before presenting it to the input of the analogue to digital converter (ADC). Next, the ADC in the acquisition system samples the signal at discrete points in time and converts the signal's voltage at these points to digital values called sample points. The horizontal system's sample clock determines how often the ADC takes a sample. The rate at which the clock "ticks" is called the sample rate and is measured in samples per second. The sample points from the ADC are stored in memory as waveform points. More than one sample point may make up one waveform point. Together, the waveform points make up one waveform record. The trigger system determines the start and stop points of the record. The display receives these record points after being stored in memory.

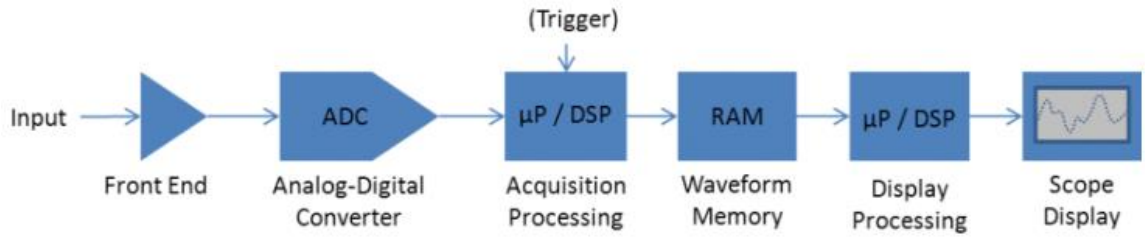


Figure 1

A function generator is usually a piece of electronic test equipment that is used to generate different types of electrical waveforms over a wide range of frequencies and amplitudes. Commonly, it is used to produce sine, square and triangular waveforms. Figure 2 illustrates the typical block diagram of function generator. A Frequency Control Network to adjust the current supplies and the frequency of the output. An integrator, or basically an operational amplifier

with a capacitor on the feedback loop. A voltage comparator that is used to form a square waveform. Resistance diode shaping circuit is used to form a sin waveform. Output amplifiers to adjust the amplitude of the waveform.

All the experiments below will be conducted using digital oscilloscope RTC 1002(figure 3a) and arbitrary function generator (AFG) HMF 2525(figure 3b) by Rodhe & Schwarz.

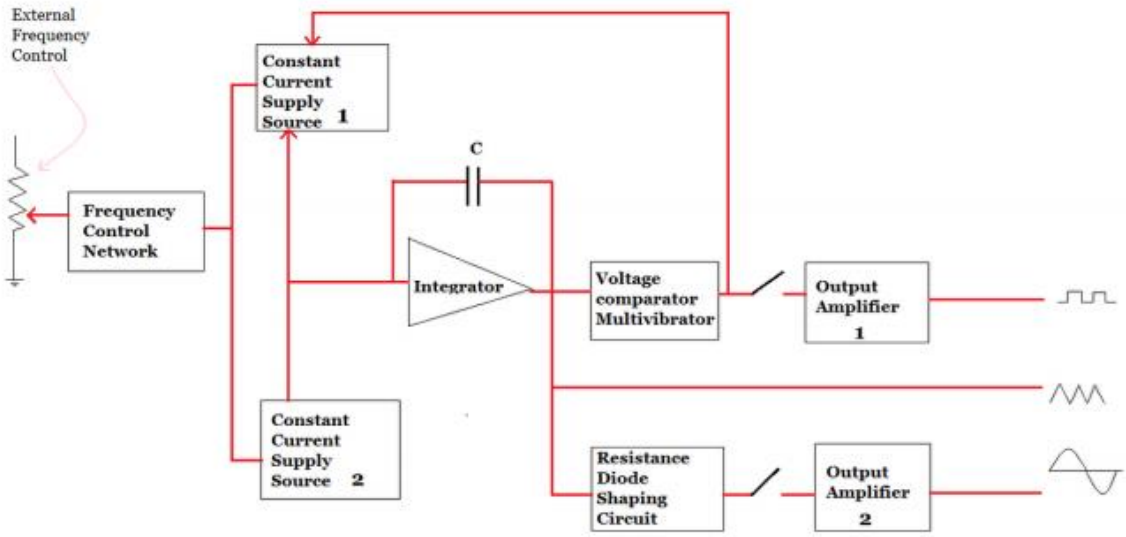


Figure 2

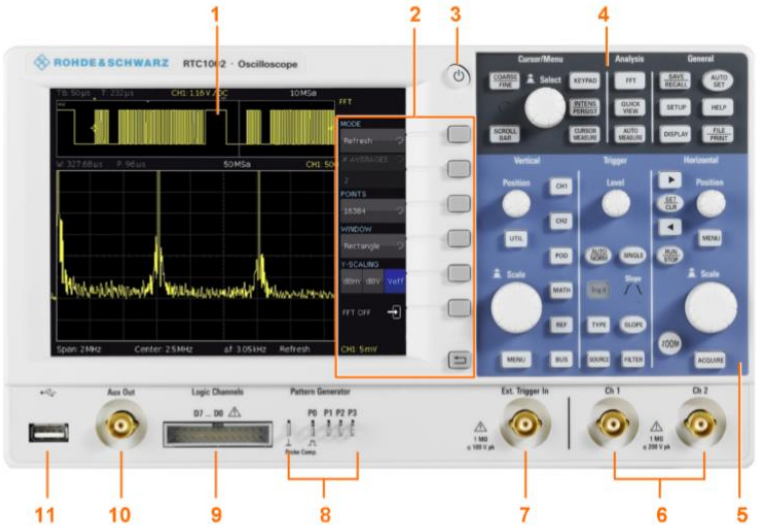


Figure 3a

1 = Display
2 = Softkeys and menu
3 = [ON/OFF] key
4 = Cursor/Menu, Analyze and General sections
5 = Vertical, Trigger and Horizontal sections
6 = BNC connectors of the analog channel inputs
7 = BNC connector of the external trigger input
8 = Pattern generator output (option R&S RTC-B6) and probe adjustment output
9 = Connector for the logic probe (option R&S RTC-B1)
10 = Multi-purpose BNC connector [Aux Out]
11 = USB connector

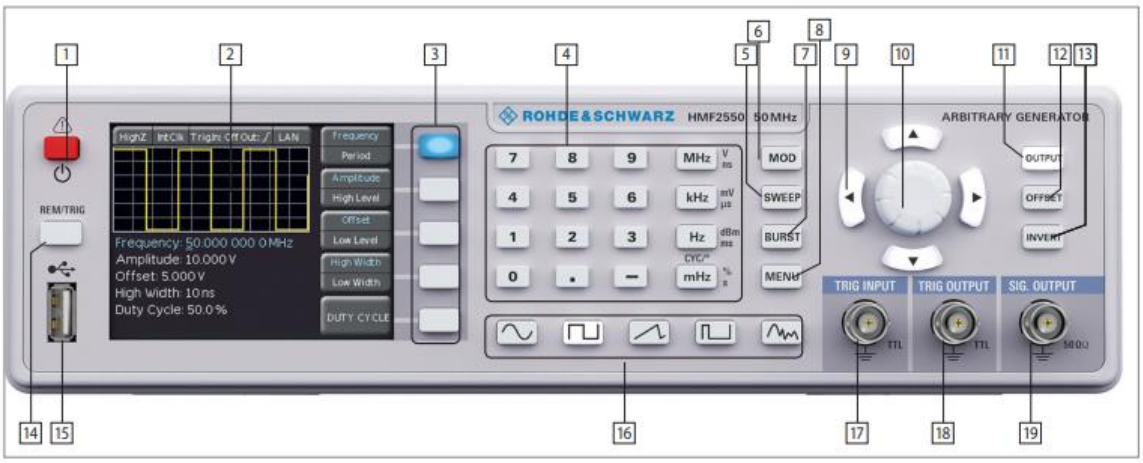


Figure 3b

1. POWER - Power switch turns the instrument on/off
2. Display (TFT) - All parameters including the current waveform are shown concurrently
3. Interactive Softkeys - Direct access of all relevant functions
4. Numerical keyboard - Setting of all operating parameters with respective units
5. SWEEP - Selection of the parameters for sweep mode
6. MOD - Modulation modes
7. BURST - Add user defined period to the waveform depending on internal or external trigger signal
8. MENU - Open the menu options
9. Arrow buttons ▲▼◄► - Cursor keys for shifting the cursor to the position to be changed, increase/decrease value of the selected parameter
10. Rotary knob - Knob to adjust the values / confirm settings by pushing the knob
11. OUTPUT - Turn on/off the output
12. OFFSET - Add a user defined DC voltage to the signal output
13. INVERT - Inverses the pulse signal output
14. REM/TRIG - Toggling between front panel and remote operation or force trigger
15. USB stick port - USB stick port for storing parameters and load waveforms
16. Signal functions - Selection of the signal: sine wave ∿, square wave ⎍, triangle ⩘, pulse ⎍, arbitrary ∿

17. TRIG INPUT (BNC socket) - Input for trigger signals
18. TRIG OUTPUT (BNC socket) - Output for trigger signals (TTL)
19. SIGNAL OUTPUT (BNC socket) - Signal output (50Ω)

Horizontal and vertical cursor measurement

Cursors are the on-screen markers associated with a channels on an oscilloscope for making measurement. Use of markers enables better accuracy than a simple grid based measurement of signal parameters. There are 2 numbers of cursors that can be moved by Universal rotary knob on the oscilloscope, very often, one by one. Cursors could be of two types – horizontal and vertical cursors. Figure 4a shows the horizontal and vertical cursors. The position of cursors can be adjusted by turning the knob. Figure 4b is the horizontal and vertical sensitivity per division. Note that vertical sensitivity scale are used to calculate the voltage/div while horizontal sensitivity are used to calculate time /division.

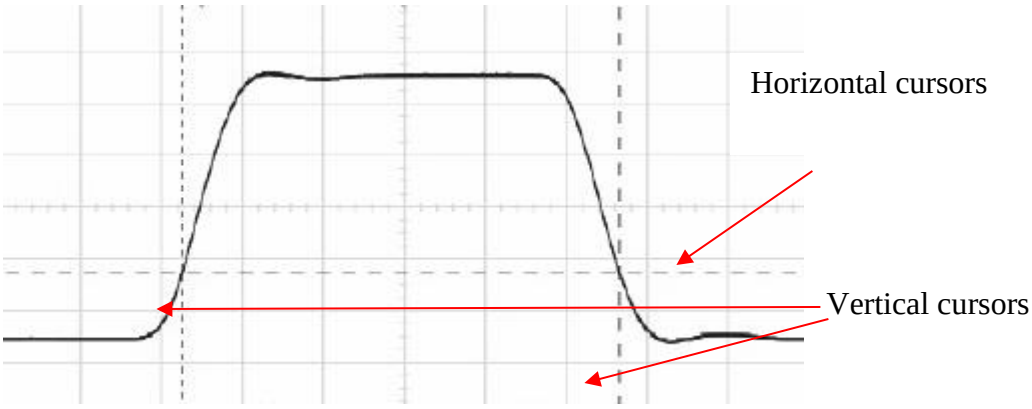


Figure 4a

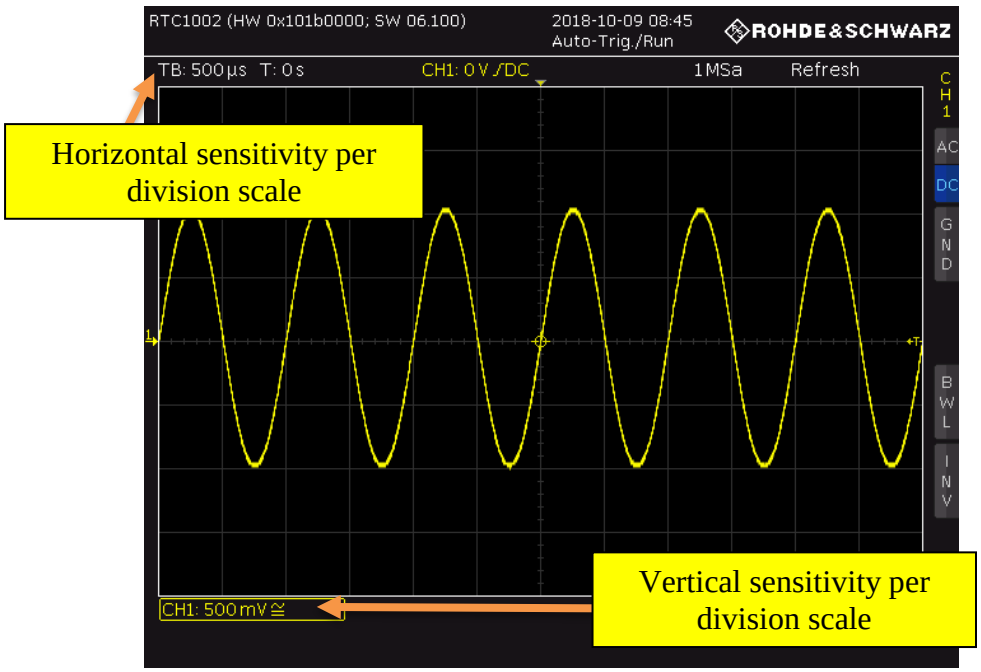


Figure 4b

The horizontal cursors are used for measurement of amplitude. The amplitude value of cursors meeting the waveform is displayed for each of the cursors – based on the waveform cursor is associated with and vertical scale of the waveform.

Signal Rise Time and Fall Time Measurement

Rise time is defined as the time it takes for a signal or rise (or fall for falltime) from 10% to 90% of its peak value. For rise time measurement, one can place two vertical cursors in such a way that it touches the 10% and 90% level of rising transition. The delta t between the cursors, measured with horizontal cursors will be the rise time. Figure 5a illustrates the rise time and fall time of a square wave signal.

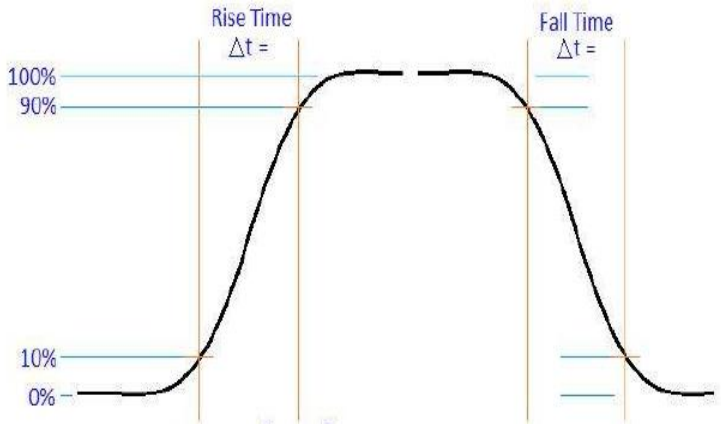


Figure 5

RMS Measurement of Signal

RMS value is a statistical measure of the magnitude of a varying quantity. RMS value of a time varying current / voltage signal is an equivalent DC signal that delivers same average power as delivered by the time varying signal. Figure 6 shows the RMS value of a sinusoidal wave.

RMS value of a sine wave = Peak Voltage / √2 (1)

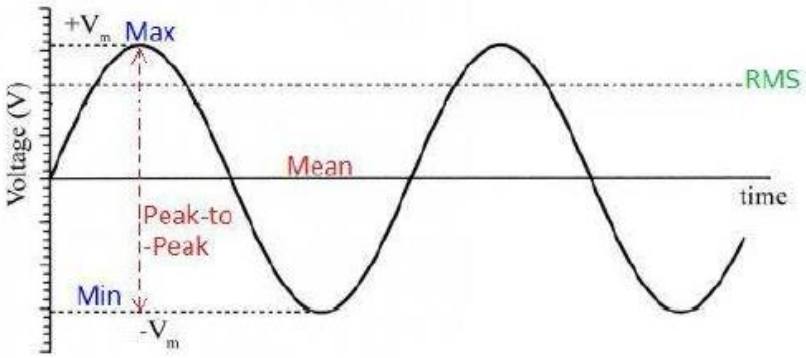


Figure 6

PART A:

AMPLITUDE AND PEAK TO PEAK SIGNAL MEASUREMENT OF AC SIGNAL

OBJECTIVE

At the end of this experiment, student will be able to measure the amplitude and peak to peak value of a signal using vertical cursors and built in function in oscilloscope

EQUIPMENT AND COMPONENT

1. Oscilloscope
2. AFG
3. BNC-BNC

PROCEDURE

1. Connect the AFG to AC power and turn ON the AFG.
2. Select the sinusoidal signal, set the frequency to 1KHz and amplitude to 1V. Refer to Figure 7 below for the signal parameter settings.



Figure 7

3. Turn ON the digital oscilloscope.
4. Connect the output of AFG to channel 1 oscilloscope by using BNC cable. Refer to Figure 8 below.



Figure 8

5. Select **CH1** to display the signal from the AFG on the oscilloscope. Refer to Figure 9.

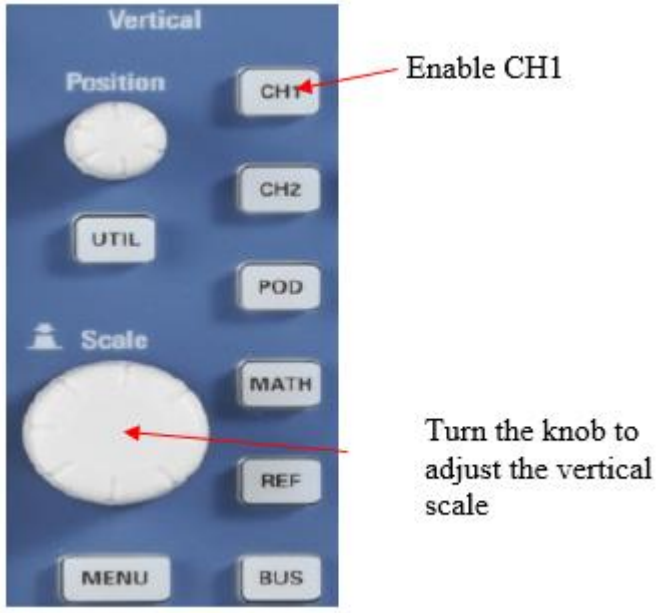


Figure 9

6. Press the output soft key in AFG to view the sinusoidal signal in oscilloscope.
7. Manually set the horizontal and vertical scale in order to view the sinusoidal signal.
8. Select and turn the **Scale** knob to select vertical scale. Refer to Figure 9. Select 200 mV/div.
9. Sketch the signal on Figure A1. Calculate your voltage readings and record in Table 1.
10. Press the **CURSOR** button at the oscilloscope panel and select **measure type: voltage**. Refer to Figure 10.



Figure 10

11. Enable **CURSOR1** and position it using the knob to touch the positive peak of the sine wave.
12. Read the display value of **CURSOR1 (V1)** and write down the value on Table 1.
13. Select **CURSOR2** and position it using the knob to touch the negative peak of the sine wave. Record down the value on V2 Table 1.
14. Read the Delta V value. What does the reading represent? Amplitude or Peak to peak voltage?
15. Go to the **Analysis** menu and select **QUICK VIEW** button on the oscilloscope front panel. Measurements will be displayed at the bottom of the screen. Look for amplitude and peak to peak voltage. Record these readings and put them in your report.
16. Press **QUICK VIEW** button again to switch off the function.
17. Change the vertical scale division to 500 mV/div.
18. Repeat step 9 – 16
19. Observe the changes and sketch the signal in Figure A2.
20. Record your observation and comparison in your report. Comment on the changes and vertical sensitivity settings.
21. Repeat the above experiment 5V signal by using 1V/div and 5V/div vertical sensitivity.
22. Sketch your waveform in Figure A3 and A4. Record your data in Table 2.

PART A RESULT:

AFG Setting: 1V

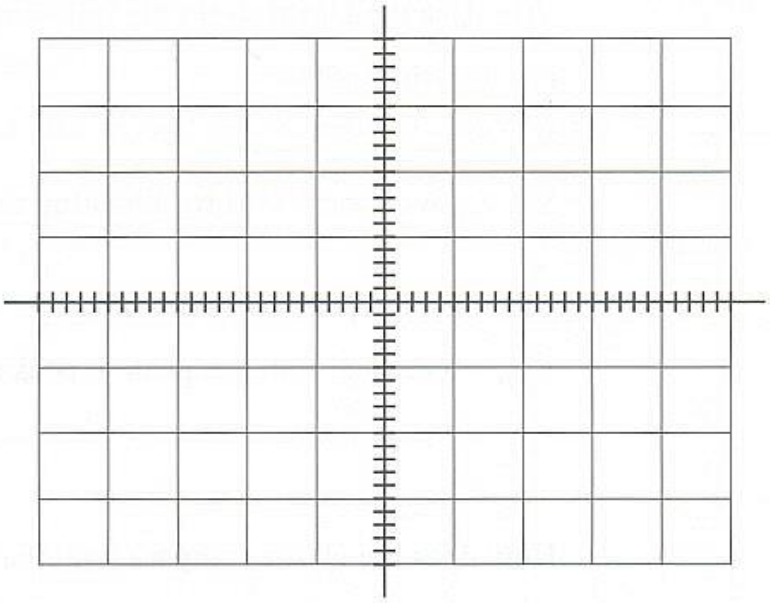


Figure A1: Wave at 200 mV/div

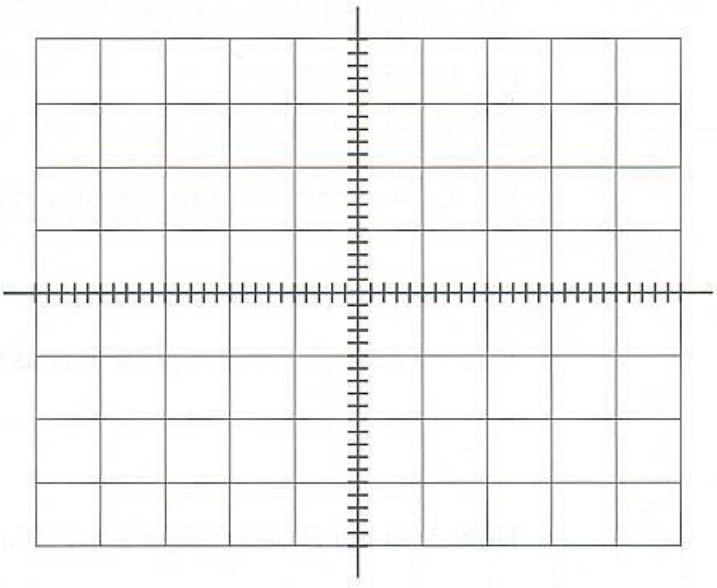


Figure A2: Waveform at 500 mV/div

Table 1: Measurement with 1V signal

	200mV/DIV			**500 mV/div**		
	Calculation	Measured (cursor)	Quick View	Calculation	Measured (cursor)	Quick View
V_1 , (V+)						
V_2, (V-)						
Delta V, Vp-p						

Note: $\boldsymbol{voltage = no\ of\ div \times vertical\ scale\ sensitivity}$

AFG Setting: 5V

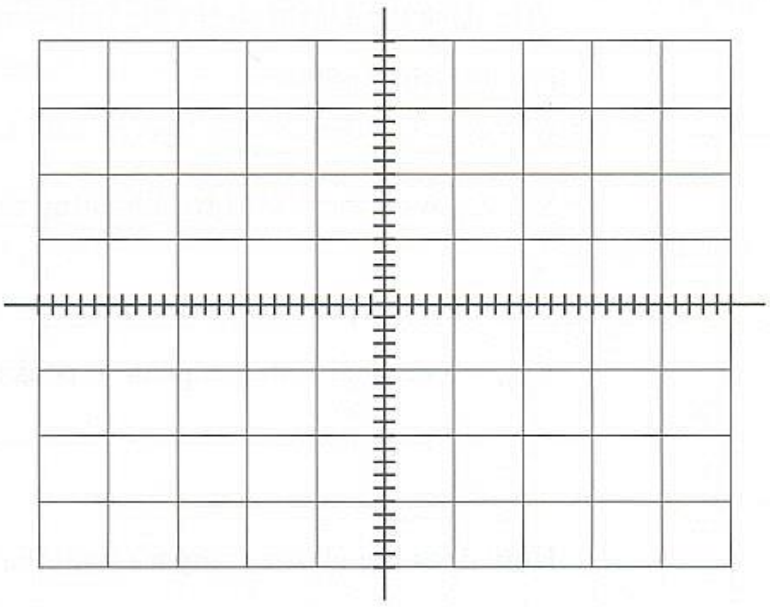


Figure A3: Wave at 1V/div

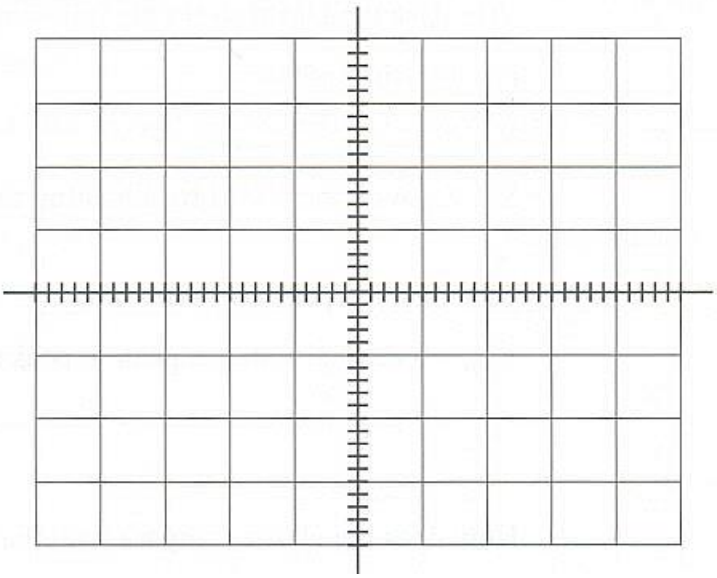


Figure A4: Waveform at 5V/div

Table2: Measurement with 5V signal

	1V/DIV			**5V/div**		
	Calculation	Measured (cursor)	Quick View	Calculation	Measured (cursor)	Quick View
V_1 , (V+)						
V_2, (V-)						
Delta V, Vp-p						

Note: $\boldsymbol{voltage = no\ of\ div \times vertical\ scale\ sensitivity}$

DISCUSSION

1. Based on your result and observation, comment on the changes and vertical sensitivity settings.
2. Briefly describe the properties of voltage peak to peak and voltage amplitude. Support your answer with relevant illustration.

PART B: PERIOD AND FREQUENCY MEASUREMENT

OBJECTIVE

At the end of this experiment, student will be able to use vertical cursors and built in functions of oscilloscope to measure the period and frequency of a signal

EQUIPMENT AND COMPONENT

1. Oscilloscope
2. AFG
3. BNC-BNC

PROCEDURE

1. Connect the AFG to AC power and turn ON the AFG.
2. Select the sinusoidal signal, set the frequency to 1KHz and amplitude to 1V.
3. Turn ON the digital oscilloscope.
4. Connect the output of AFG to channel 1 oscilloscope by using BNC cable.
5. Select CH1 to display the signal from the AFG on the oscilloscope.
6. Press the output soft key in AFG to view the sinusoidal signal in oscilloscope.
7. Manually set the horizontal and vertical scale as well as the triggering in order to view the sinusoidal signal.
8. Adjust the vertical scale sensitivity to 500 mV/div
9. Adjust the horizontal scale by turning the **Scale** knob as shown in Figure 11 below. Adjust until you see 6 cycles of signal. Sketch the signal on Figure B1.
10. What is the horizontal scale sensitivity? Record your value in result.

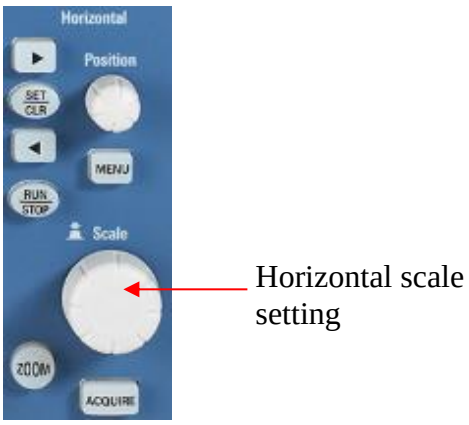


Figure 11

11. Calculate frequency and period (T). Record in Table 3.
12. Go to the cursor menu by pressing **CURSOR** button on the oscilloscope front panel. Select the **measure type: time**. Place **CURSOR1** and **CURSOR 2** at the positions as shown in Figure 12.

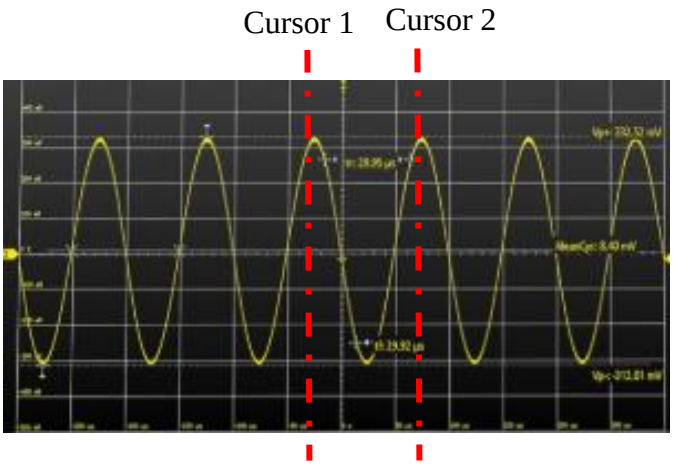


Figure 12

13. Record the delta t value and the frequency value in Table 3.
14. Go to the **Analysis** menu and select **QUICK VIEW** button on the oscilloscope front panel. Measurements will be displayed at the bottom of the screen. Look for period and frequency. Record these readings and put them in your report.
15. Compare the reading obtained from procedure 10 and procedure 14. If there is difference, write down your comment in your report.
16. Turn the knob until you see 2 cycles of signal. What is the horizontal scale sensitivity? Record your calculation and observation in Figure B2 and Table 3.
17. Repeat the above experiment for sinusoidal signal (except step 9 and 16) with:
    (i) frequency=2 KHz and
    (ii) frequency=8 KHz.

    NOTE: USE SAME HORIZONTAL SCALE SENSITIVITY AS PROCEDURE 10 AND 16.

RESULT

Frequency setting: 1KHz

Horizontal Scale Sensitivity 1: _____

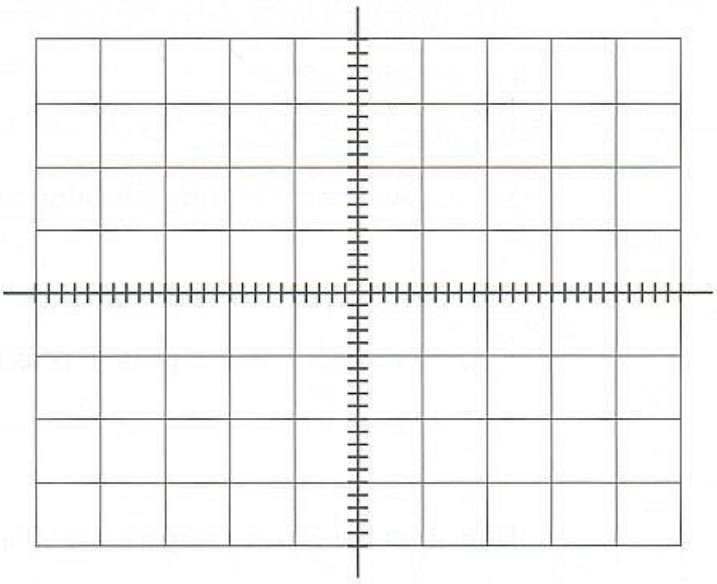


Figure B1: Waveform at horizontal scale sensitivity 1 at 1KHz

Horizontal Scale Sensitivity 2: _____

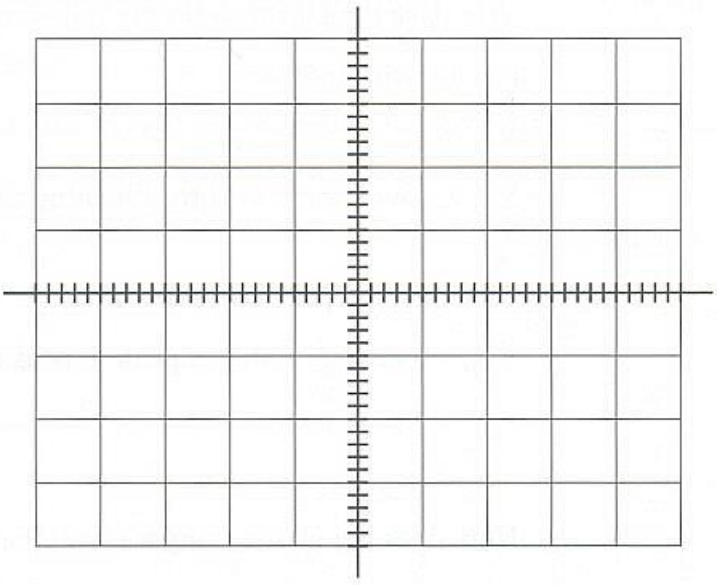


Figure B2: Waveform at horizontal scale sensitivity 2 at 1KHz

Table 3: Measurement with 1KHz frequency signal

	Horizontal Scale Sensitivity 1			Horizontal Scale Sensitivity 2		
	Calculation	Measured (cursor)	Quick View	Calculation	Measured (cursor)	Quick View
Period, T						
Frequency						

Frequency setting: 2KHz

Horizontal Scale Sensitivity 1: _____

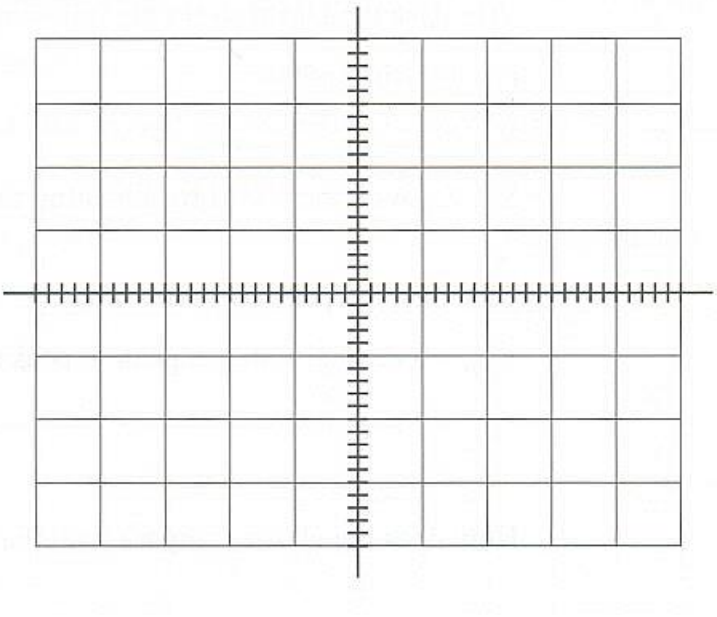


Figure B3: Waveform at horizontal scale sensitivity 1 at 2KHz

Horizontal Scale Sensitivity 2: _____

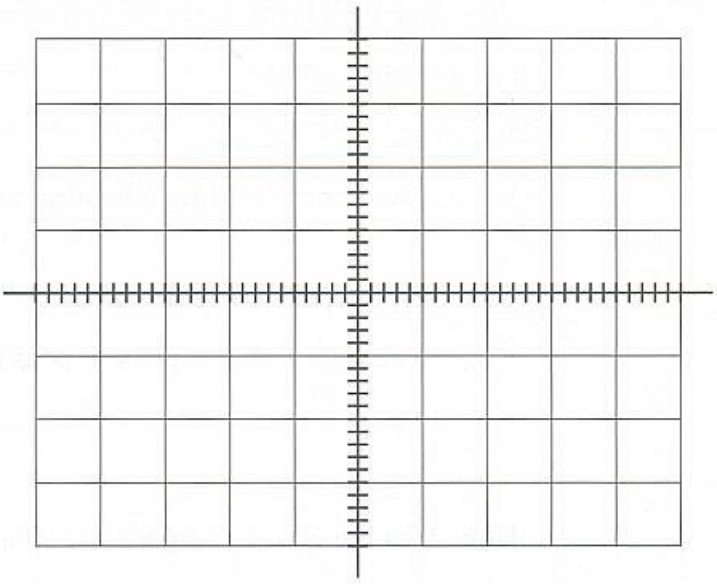


Figure B4: Waveform at horizontal scale sensitivity 2 at 2KHz

Table 4: Measurement with 2KHz frequency signal

	Horizontal Scale Sensitivity 1			Horizontal Scale Sensitivity 2		
	Calculation	Measured (cursor)	Quick View	Calculation	Measured (cursor)	Quick View
Period, T						
Frequency						

Frequency setting: 8KHz

Horizontal Scale Sensitivity 1: _____

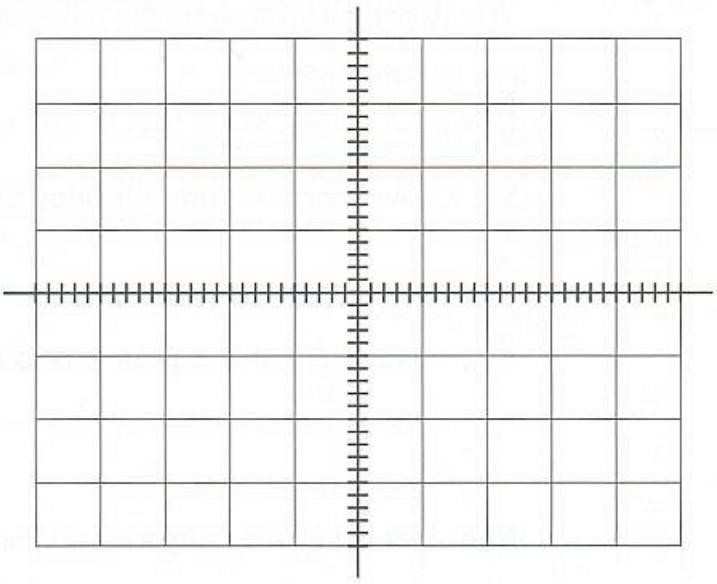


Figure B5: Waveform at horizontal scale sensitivity 1 at 8KHz

Horizontal Scale Sensitivity 2: _____

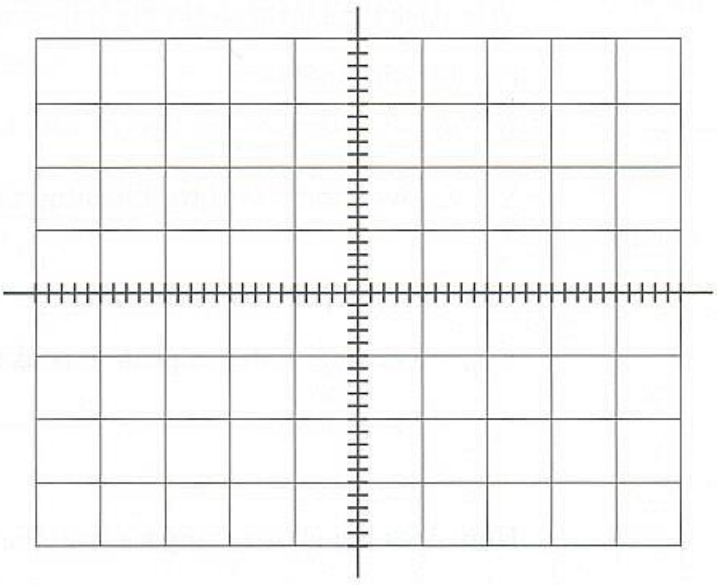


Figure B6: Waveform at horizontal scale sensitivity 2 at 8KHz

Table 5: Measurement with 8KHz frequency signal

	Horizontal Scale Sensitivity 1			Horizontal Scale Sensitivity 2		
	Calculation	Measured (cursor)	Quick View	Calculation	Measured (cursor)	Quick View
Period, T						
Frequency						

DISCUSSION

1. Based on your result and observation, comment on the changes and horizontal sensitivity settings.
2. Briefly describe the properties of frequency and time period. Support your answer with relevant illustration.

PART C: RMS MEASUREMENT OF SIGNAL

OBJECTIVE

At the end of this experiment, student will be able to measure the RMS value of a signal using built in function in oscilloscope

EQUIPMENT

1. Oscilloscope
2. AFG
3. BNC-BNC

PROCEDURE

1. Connect the AFG to AC power and turn ON the AFG.
2. Select the sinusoidal signal, set the frequency to 1KHz and amplitude to 1V.
3. Turn ON the digital oscilloscope.
4. Connect the output of AFG to channel 1 oscilloscope by using BNC cable.
5. Select CH1 to display the signal from the AFG on the oscilloscope.
6. Press the output soft key in AFG to view the sinusoidal signal in oscilloscope.
7. Manually set the horizontal and vertical scale as well as the triggering in order to view the sinusoidal signal.
8. Go to the **Analysis** menu and select **QUICK VIEW** button on the oscilloscope front panel. Measurements will be displayed at the bottom of the screen. Look for RMS reading. Record the reading and put it in your report.
9. Calculate the theoretical RMS value using Equation 1. Compare and comment on the reading from procedure 6 and procedure 7.
10. Sketch your waveform with appropriate scaling and parameter in Figure C1.

RESULT

Table 6: RMS Measurement

	Calculation	Quick View
V_{Amp}		
$V_{peak\text{-}peak}$		
V_{RMS}		

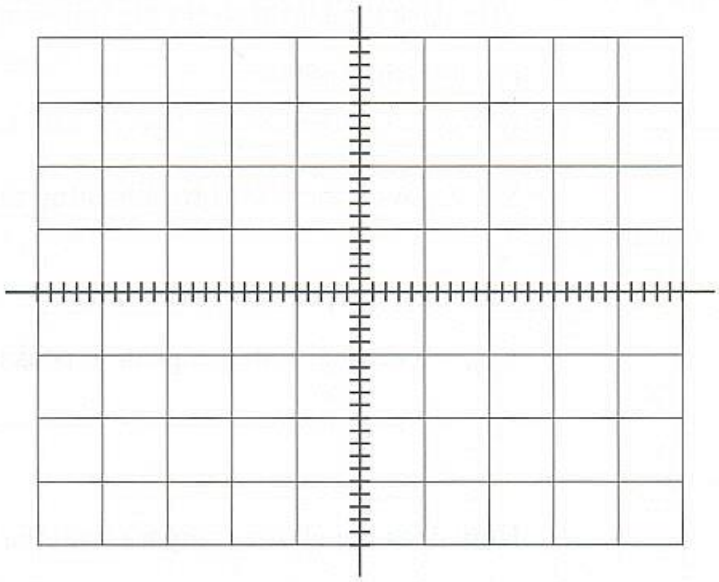


Figure C1

DISCUSSION

1. Based on your result and observation, comment on your result.

PART D: SIGNAL RISE TIME AND FALL TIME MEASUREMENT

OBJECTIVE

At the end of this experiment, student will be able to measure rise and fall time of a signal using horizontal and vertical cursors and the built in function in oscilloscope.

EQUIPMENT

4. Oscilloscope
5. AFG
6. BNC-BNC

PROCEDURE

1. Connect the AFG to AC power and turn ON the AFG.
2. Select the square waveform signal. Refer to Figure 13 below. Set the frequency to 1KHz and amplitude to 1V.



Figure 13

3. Turn ON the digital oscilloscope.
4. Connect the output of AFG to channel 1 oscilloscope by using BNC cable.
5. Select **CH1** to display the signal from the AFG on the oscilloscope.
6. Press the output soft key in AFG to view the square wave signal in oscilloscope.
7. Manually set the horizontal and vertical scale as well as the triggering in order to view the square wave signal.

8. Adjust the vertical scale sensitivity to 200 mV/div. Sketch the square wave signal on Figure D1.
9. Adjust the vertical position so the step amplitude is exactly 5 vertical divisions. Each vertical division will represent 20% of the step amplitude.
10. Press slope. Make sure rise slope is selected.
11. Press ZOOM button.
12. Adjust the time/horizontal scale and horizontal position using the knob to zoom in at the rising edge of the pulse.

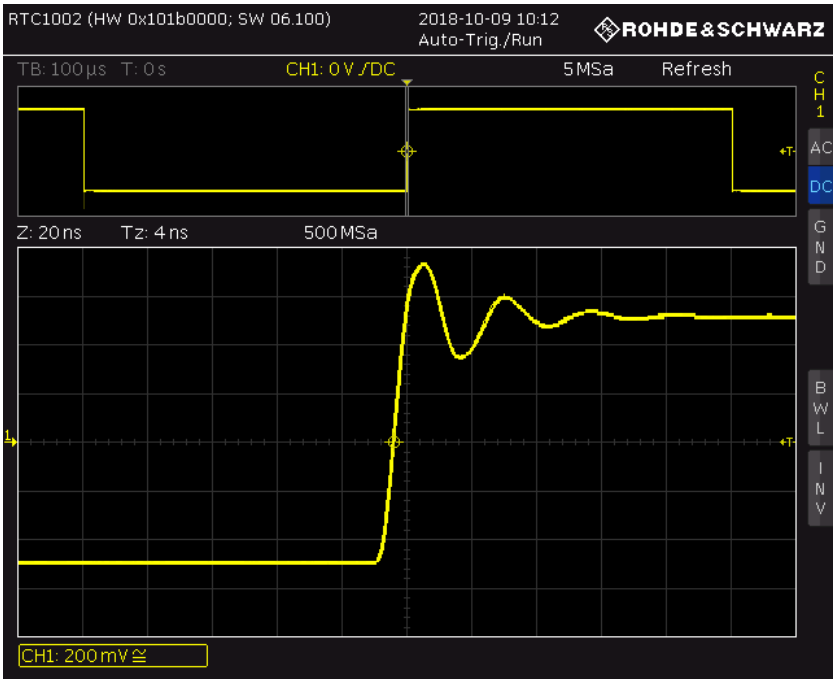


Figure 13

13. Press cursor measure. Select measure type time.
14. Set your vertical cursors to obtain the rise time.
15. **CURSOR1** at 10% of the peak amplitude. Record your value.
16. **CURSOR2** at 90% of the peak amplitude. Record your value.
17. Read the **Delta** value and put it in your report.
18. Repeat the procedure 1 - 9 to obtain the fall time of the signal.
19. Press slope. Make sure fall slope is selected.
20. Press ZOOM button.

21. Adjust the time/horizontal scale and horizontal position using the knob to zoom in at the falling edge of the square wave to read the fall time.
22. Set your vertical cursors to obtain the fall time.
23. **CURSOR1** at 10% of the peak amplitude. Record your value.
24. **CURSOR2** at 90% of the peak amplitude. Record your value.
25. Read the **Delta V** value and put it in your report.
26. Disable zoom function.
27. Go to the **Analysis** menu and select **QUICK VIEW** button on the oscilloscope front panel. Measurements will be displayed at the bottom of the screen. Look for rise time and fall time readings. **You may need to adjust the horizontal scale to get an accurate value**. Record these readings and put them in your report. Compare and comment on the rise time and fall time obtained from procedure 6 and procedure 8.

RESULT

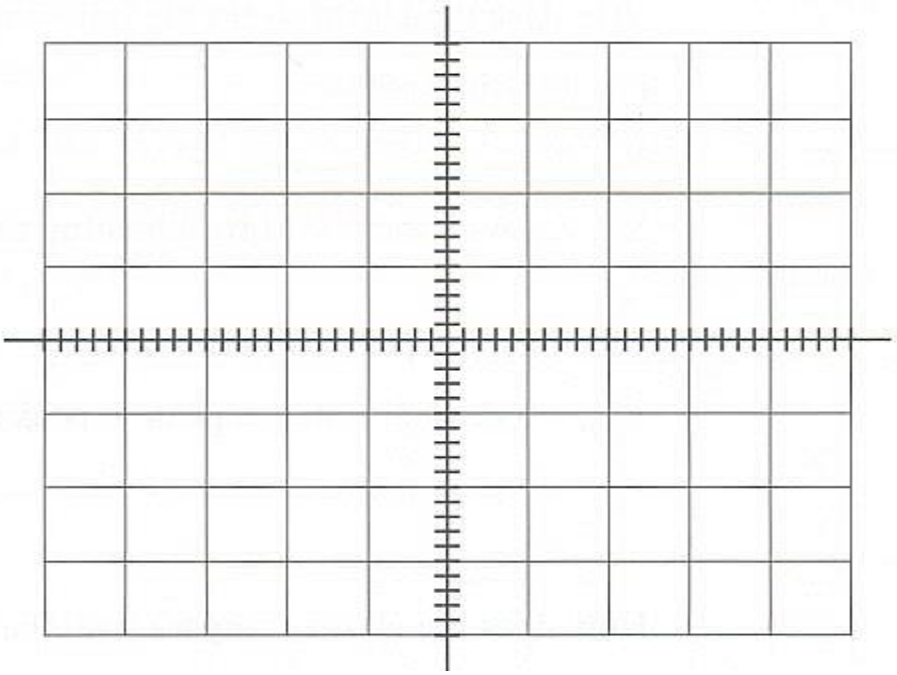


Figure D1: Rise Time

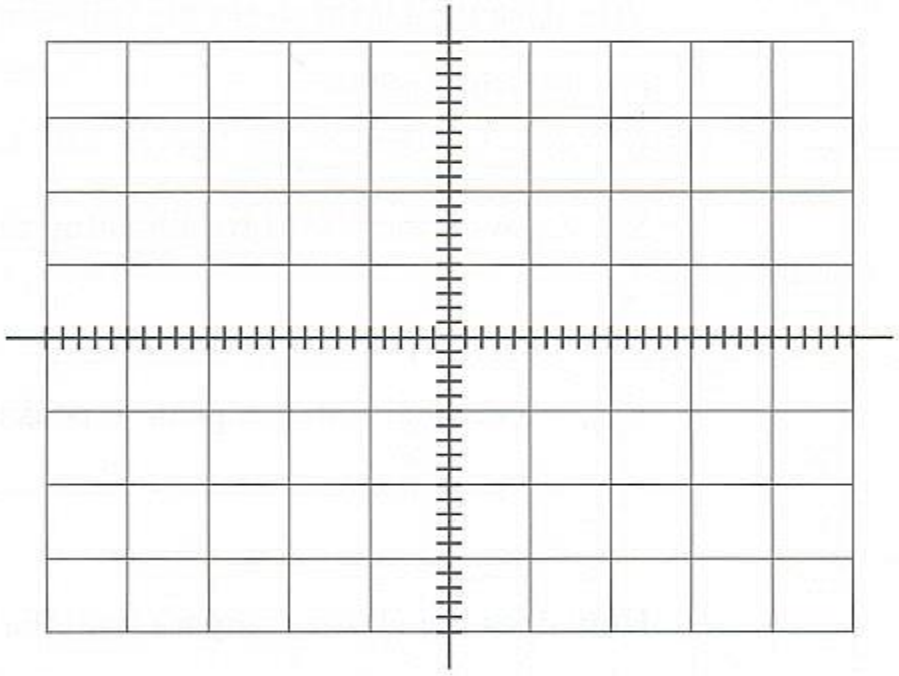


Figure D2: Fall Time

Table 7: Rise time and fall time measurement

	Rise Time		Fall Time	
	Cursor	Quick View	Cursor	Quick View
t_1				
t_2				
Δ				

DISCUSSION

1. Based on your result and observation, comment on your result.